```
import tensorflow as tf
from tensorflow.keras.preprocessing.image import ImageDataGenerator
from tensorflow.keras.applications import MobileNetV2
from tensorflow.keras.layers import GlobalAveragePooling2D, Dense
from tensorflow.keras.models import Sequential
import zipfile
import os
```

```
from google.colab import files

uploaded = files.upload()
```

Choose Files 30eebb98-c…8IDTaM.zip

30eebb98-ccbe-4fd3-b844-38660711c428IDTaM.zip(application/x-zip-compressed) - 15828590 bytes, last modified: 7/3/2026 - 100% done

```
Saving 30eebb98-ccbe-4fd3-b844-38660711c428IDTaM.zip to 30eebb98-ccbe-4fd3-b844-38660711c428IDTaM.zip
```

```
zip_file = list(uploaded.keys())[0]

with zipfile.ZipFile(zip_file, 'r') as zip_ref:
    zip_ref.extractall("BrainMRI")

print("Dataset Extracted Successfully!")
```

```
Dataset Extracted Successfully!
```

```
train_dir = "BrainMRI/brain_tumor_dataset"
test_dir = "BrainMRI"
```

```
train_datagen = ImageDataGenerator(
    rescale=1./255,
    rotation_range=20,
    zoom_range=0.2,
    horizontal_flip=True
)

test_datagen = ImageDataGenerator(
    rescale=1./255
)
```

```
train_generator = train_datagen.flow_from_directory(
    train_dir,
    target_size=(224,224),
    batch_size=32,
    class_mode='binary'
)

test_generator = test_datagen.flow_from_directory(
    test_dir,
    target_size=(224,224),
    batch_size=32,
    class_mode='binary'
)
```

```
Found 253 images belonging to 2 classes.
Found 506 images belonging to 3 classes.
```

```
base_model = MobileNetV2(
    weights='imagenet',
    include_top=False,
    input_shape=(224,224,3)
)
base_model.trainable = False
```

```
Downloading data from https://storage.googleapis.com/tensorflow/keras-applications/mobilenet_v2/mobilenet_v2_weights_tf_dim_order
9406464/9406464 ━━━━━━━━━━━━━━━━━━━━ 0s 0us/step
```

```
model = Sequential([
    base_model,
    GlobalAveragePooling2D(),
```

```
    Dense(1, activation='sigmoid')
])
model.compile(
    optimizer='adam',
    loss='binary_crossentropy',
    metrics=['accuracy']
)
model.summary()
```

Model: "sequential"

Layer (type)	Output Shape	Param #
mobilenetv2_1.00_224 (Functional)	(None, 7, 7, 1280)	2,257,984
global_average_pooling2d (GlobalAveragePooling2D)	(None, 1280)	0
dense (Dense)	(None, 1)	1,281

Total params: 2,259,265 (8.62 MB)
Trainable params: 1,281 (5.00 KB)